



AQ9.5: Activity Questions 5 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

1 point

Let $f(x, y, z) = (xy, x + y + z^2)$, where $x, y, z \in \mathbb{R}$. Choose the set of correct options.

☐

The domain of f is $\mathbb{R}^2$ and the co-domain of f is $\mathbb{R}^2$

☐

The domain of f is $\mathbb{R}^3$ and the co-domain of f is $\mathbb{R}^2$

☐

f is continuous everywhere in its domain

☐

f is not continuous anywhere in its domain

No, the answer is incorrect.

Score: 0

Accepted Answers:

The domain of f is $\mathbb{R}^3$ and the co-domain of f is $\mathbb{R}^2$

f is continuous everywhere in its domain

1 point

Let $f(x, y) = \begin{cases} \frac{\cos y \sin x}{\cos x} & x \neq 0 \\ \sin y & x = 0 \end{cases}$, $f(x, y) = \begin{cases} x \cos y \sin x & x \neq 0 \\ 0 & x = 0 \end{cases}$.

Choose the correct options.

☐

f is not defined at $(0, 0)$

☐

f is defined at $(0, 0)$

☐

f is continuous at $(0, 0)$

☐

f is not continuous at $(0, 0)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

f is defined at $(0, 0)$

f is continuous at $(0, 0)$

1 point

Let $h(u, v) = \sin(uv)$, $g(x) = x^3$, $f(y) = \cos y$, $h(u, v) = \sin(uv)$, $g(x) = x^3$, $f(y) = \cos y$. Let $p(x, y) = h(g(x), f(y))$

Choose the correct option(s).

☐

p is a scalar-valued function

☐

p is a vector-valued function

☐


pp is continuous everywhere since $g.f$ $g.f$ is continuous everywhere in its domain

☐

pp is continuous everywhere, gg is also continuous everywhere while ff is not continuous everywhere in its domain

No, the answer is incorrect.

Score: 0

Accepted Answers:

pp is a scalar-valued function

pp is continuous everywhere since $g.f$ $g.f$ is continuous everywhere in its domain

1 point

Let $p(t) = (\ln(1 - t), \frac{1}{t}, 3t)$ $p(t) = (\ln(1 - t), t1, 3t)$.

Choose the incorrect option(s).

☐

pp is continuous for all real t less than 11 but not equal to zero

☐

pp is continuous on $\mathbb{R}$

☐

pp is continuous in $\{t \in \mathbb{R} | t < 1\}$ $\{t \in \mathbb{R} | t < 1\}$

☐

pp is continuous in $\{t \in \mathbb{R} | t \neq 1, t \neq 0\}$ $\{t \in \mathbb{R} | t = 1, t = 0\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

pp is continuous on $\mathbb{R}$

pp is continuous in $\{t \in \mathbb{R} | t < 1\}$ $\{t \in \mathbb{R} | t < 1\}$

pp is continuous in $\{t \in \mathbb{R} | t \neq 1, t \neq 0\}$ $\{t \in \mathbb{R} | t = 1, t = 0\}$

1 point

Let $f(x, y) = \ln(x^2 + y^2 - 1)$ $f(x, y) = \ln(x^2 + y^2 - 1)$. Identify where f is continuous. Note:

Equation of a circle with radius r and center (a, b) is given by $(x - a)^2 + (y - b)^2 = r^2$

$(x - a)^2 + (y - b)^2 = r^2$

☐

Everywhere in $\mathbb{R}^2$

☐

Everywhere in $\mathbb{R}^2 \setminus \{(1, 0), (0, 1)\}$ $\mathbb{R}^2 \setminus \{(1, 0), (0, 1)\}$

☐

Everywhere in $\mathbb{R}^2$ outside the circle with radius 1 and center $(0, 0)$

☐ None of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

Everywhere in $\mathbb{R}^2$ outside the circle with radius 1 and center $(0, 0)$

Level 2

1 point

Let $g(u, v, w) = \frac{uv}{w}$ $g(u, v, w) = wuv$ and $f(x, y, z) = g(P(x, y, z), Q(x, y, z), R(x, y, z))$

$f(x, y, z) = g(P(x, y, z), Q(x, y, z), R(x, y, z))$

where $P(x, y, z) = e^{x^2 + y}$, $Q(x, y, z) = \sqrt{y^2 + z^2 + 3}$, $R(x, y, z) = \sin(xyz) + 5$

$$P(x, y, z) = ex^{2+y}, Q(x, y, z) = y^2 + z^2 + 3$$

$$\sqrt{}, R(x, y, z) = \sin(xyz) + 5.$$

Choose the correct option(s).

☐

f is a vector-valued function

☐

f is a scalar-valued function

☐

f is continuous everywhere in its domain.

☐

P is continuous everywhere in its domain, while Q is not continuous everywhere in its domain

No, the answer is incorrect.

Score: 0

Accepted Answers:

f is a scalar-valued function

f is continuous everywhere in its domain.

1 point

Let

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x, y) = \begin{cases} \frac{x^2}{x^2 + y^2}(x, y) & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases} \quad f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x, y) = \begin{cases} x^2 + y^2 x y 0(x, y) & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

Choose the correct option(s).

☐

$\lim_{(x, y) \rightarrow (0, 0)} f$ does not exist

☐

$\lim_{(x, y) \rightarrow (0, 0)} f$ exists

☐

f is continuous at $(0, 0)$

☐

f is not continuous at $(0, 0)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\lim_{(x, y) \rightarrow (0, 0)} f$ does not exist

f is not continuous at $(0, 0)$

1 point

Let

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}(x, y) & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases} \quad f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x, y) = \begin{cases} x^2 + y^2 xy 0(x, y) & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

Choose the correct option(s).

☐

$\lim_{(x, y) \rightarrow (0, 0)} f$ does not exist

☐

$\lim_{(x,y) \rightarrow (0,0)} f$ exists

☐

f is continuous at $(0,0)$

☐

f is not continuous at $(0,0)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\lim_{(x,y) \rightarrow (0,0)} f$ does not exist

f is not continuous at $(0,0)$

1 point

Let $f(r) = \frac{1}{\langle r, r \rangle - 1}$, where $r = (x, y, z)$ and the inner product $\langle, \rangle$ is the dot product
Choose the set of incorrect options.

☐

f is continuous throughout $\mathbb{R}^3$

☐

f is continuous throughout $\mathbb{R}^3 \setminus \{(0, 0, 0)\}$

☐

f is discontinuous on the unit sphere $\{r \in \mathbb{R}^3 : \|r\| = 1\}$

☐

f is continuous throughout $\mathbb{R}^3 \setminus \{(1, 0, 0)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

f is continuous throughout $\mathbb{R}^3$

f is continuous throughout $\mathbb{R}^3 \setminus \{(0, 0, 0)\}$

f is continuous throughout $\mathbb{R}^3 \setminus \{(1, 0, 0)\}$

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = \begin{cases} \frac{x^k y}{x^4 + y^4} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$

$(0, 0)$. Find the minimum possible positive integer value of k such that f is continuous at $(0, 0)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 4

1 point

[Check Answers](#)

Your score is: 0/10